Week 2

Review paper discussion and group/research question

Group discussions

- In 4 groups, discuss the review paper(~20 minutes)
- Points to discuss:
  - What is the structure of the paper?
  - What is the research question of this paper?
  - What, other than ML methods, is presented?
  - How does the paper present and compare the different methods?
  - How many references does the paper have?
  - What are the strong / weak points of this review?
- Class discussion (5-10 min)

Choose literature & collect relevant material

How to find sources?

- How do you do state of the art research?

Look for keywords

- Determine what keywords define your technology and domain
  - Also find synonyms
- Search for single keywords and combinations
  - Transformers
  - Transformer models
  - Solar activity prediction
  - Transformers for Solar activity prediction
  - Etc.

Literature research search

- Search engines
  - Will find everything, good and bad
- Github etc.
  - Can be used to search trending repositories in ML
- Specialized search engines
  - Google Scholar, ResearchGate etc.
    - Allows to easily find related publications
  - Arxiv
    - Not always reviewed!
  - Papers with code
    - Great resource for reproducible science papers
- Journals / Conference websites
  - Specialized on a specific domain

Literature research sources

- Blogs
  - Personal opinions, not reviewed, can still be useful
- Wikipedia, specialized websites
  - While not properly reviewed, often give good insight and further resources on specific subjects
- Pre-print research papers
  - The cutting edge of research, but not yet reviewed. Can be months ahead of properly reviewed research
- Journal and conference papers
  - Reviewed research
  - Not all journals and conferences are of equal quality
  - Number of citations is an indication (but no proof) of a quality research paper
    - Not really applicable for cutting edge research

Tips and tricks

- Use sites like google scholar, research gate and others to see how many citations a paper has
- The HES-SO and other universities have subscriptions to certain journals like IEEE for example
  - You need to be in the network/VPN to get access
- If you can't get the PDF, try googling the name
  - Authors often upload their publications on arxiv, research gate and others
- Some publications come from conferences and have videos
- ChatGPT/Gemini etc. are a good start to find important papers and keywords
  - But really unreliable and incomplete, double check everything and do your own research

Follow your sources

- During your search you will find sources do not read them all but try to follow them
  - They reference other sources
  - They contain new keywords you might not have thought about
- This can quickly get out of hand, stay focused on your research question

How do i keep track?

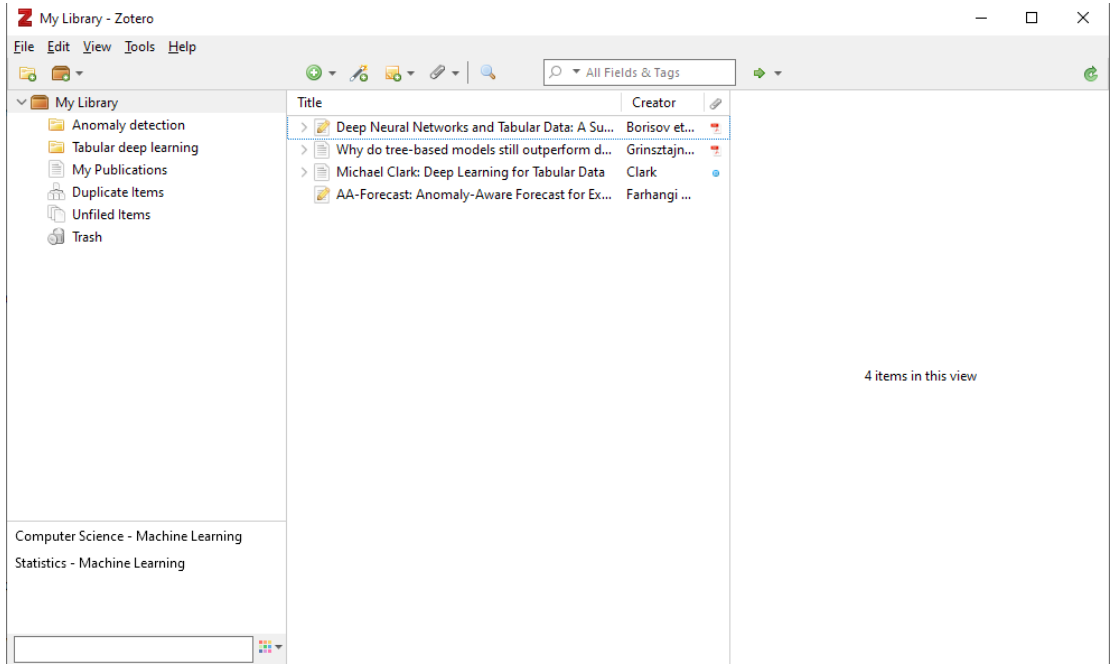
- Doing literally research can quickly be overwhelming.
  - Too much information in too many places
- Tools like Mendeley or Zotero help you to stay organized
- They allow you to collect, tag, organise, group etc. research you find
- They also make it easy to create a bibliography

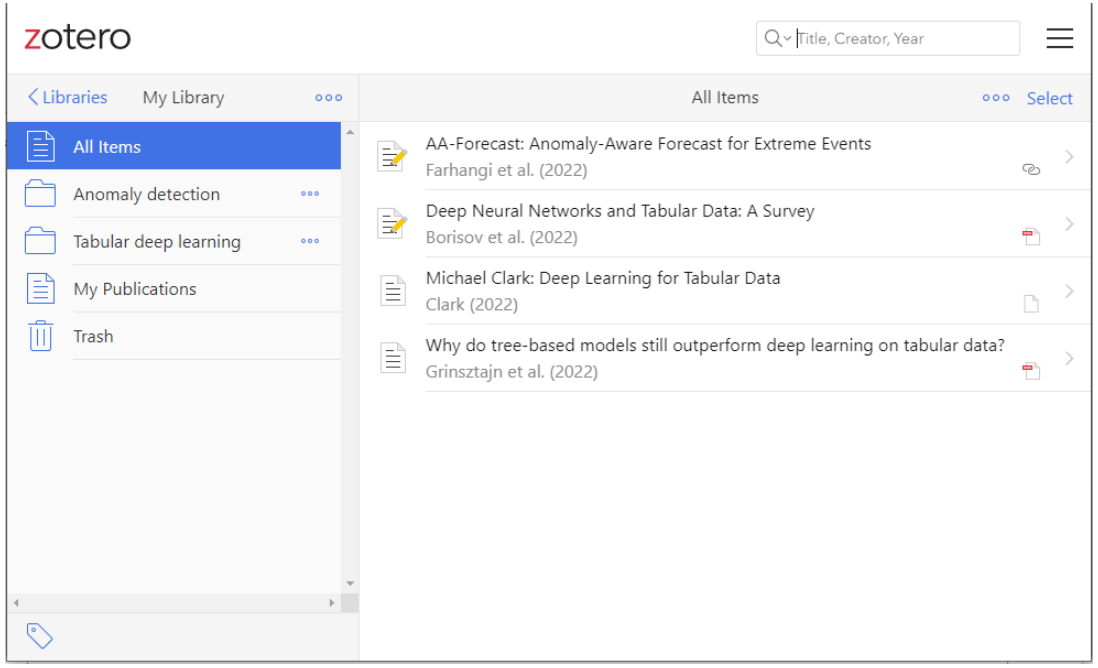




Zotero

- Easy to use desktop and web application (synced)
  - Allows to share research papers in groups
- There are also mobile applications

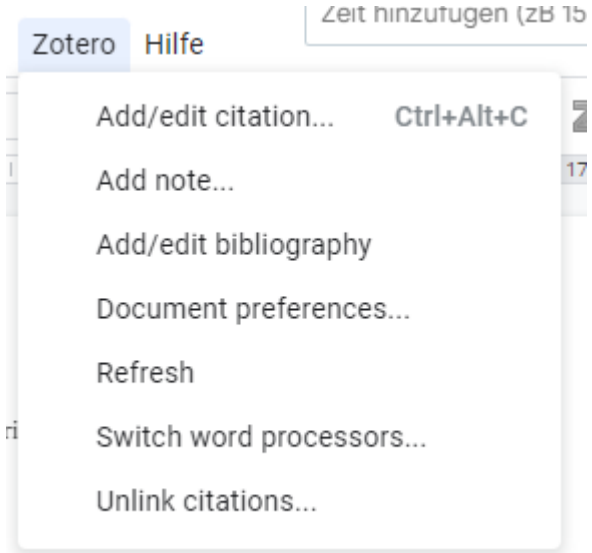


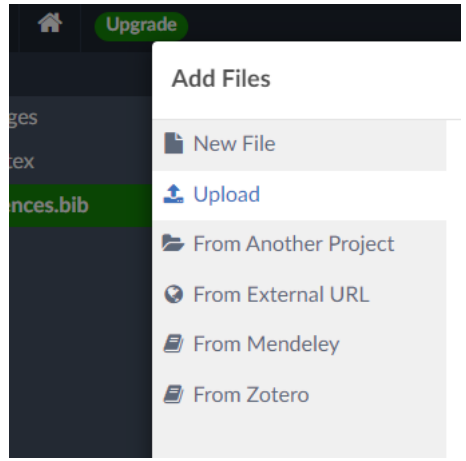


Zotero - Integrations

- Web-browser integration
  - Fast transfer of publications, websites etc into your collection
- Integration Word, Google docs, Overleaf
  - Easy integration of references into your documents







Results

- For your research question you have an extensive bibliography
  - Which you did not read yet (or just partially)
- This literature research can be too extensive and include less reputable sources
  - You can filter later
- While you are not supposed to read it all, maybe look at some sources that seem highly relevant
  - Helps you to better understand the subject, find more keywords and sources

Read/skim articles & Group them by theme

How do i read so much?

- You do not need to read every publication, blogpost etc. in detail
- Reading the title, abstract and maybe conclusion will give you a quick idea
  - You can read the details later for the sources that seem most interesting
- You are not trying to make a complete list of sources covering your research question
  - Focus on the most important ones
- Group your sources by theme
  - This will help you to stay coherent

Read quickly

- To read quickly, follow this common strategy and read:
  - Title, author, journal, date
  - Abstract
  - Table of contents
  - Introduction + conclusion
  - Figures
  - First sentence of each paragraph
- Knowing this will also help you to write better research articles

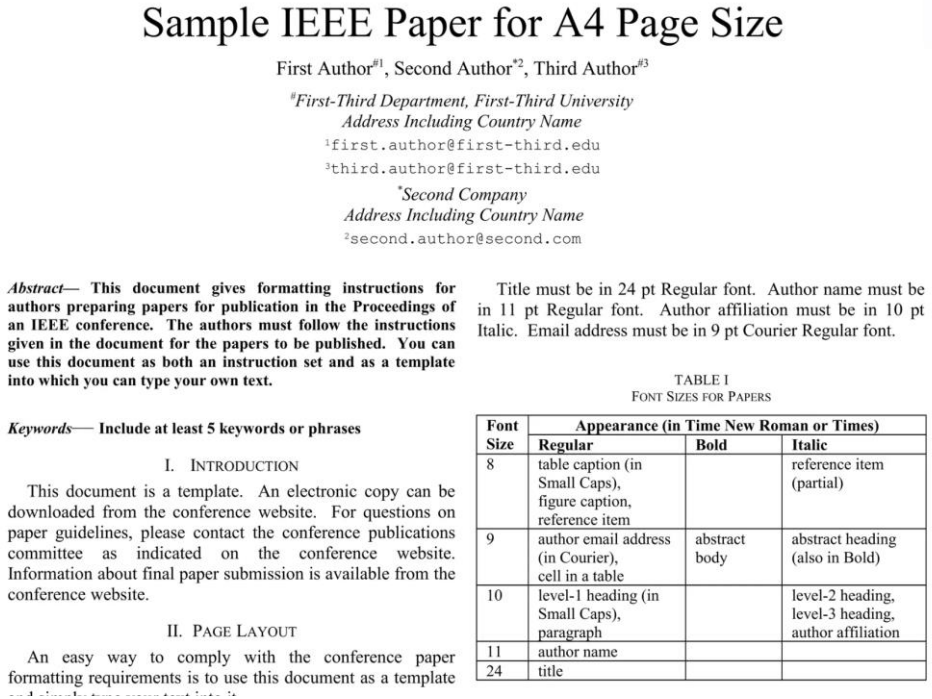
Create notes

- While reading, make small notes
  - Or highlight parts etc.
- This will make the analyzing and writing phase much faster

Analyze and write

Short paper

- Different standards exist, we use:
  - IEEE 8 pages, 2 column, figures in columns
- Standard format with 2 columns of text per page
- You can easily find Latex and Word templates
  - https://www.overleaf.com/latex/templates/ieee-conference-template/grfzhhncsfqn



Sample IEEE Paper for A4 Page Size

First Author[#1], Second Author[*2], Third Author[#3]

[#]First-Third Department, First-Third University
Address Including Country Name
[1]first.author@first-third.edu
[3]third.author@first-third.edu

[*]Second Company
Address Including Country Name
[2]second.author@second.com

***Abstract*— This document gives formatting instructions for authors preparing papers for publication in the Proceedings of an IEEE conference. The authors must follow the instructions given in the document for the papers to be published. You can use this document as both an instruction set and as a template into which you can type your own text.**

***Keywords*— Include at least 5 keywords or phrases**

I. Introduction

This document is a template. An electronic copy can be downloaded from the conference website. For questions on paper guidelines, please contact the conference publications committee as indicated on the conference website. Information about final paper submission is available from the conference website.

II. Page Layout

An easy way to comply with the conference paper formatting requirements is to use this document as a template

Title must be in 24 pt Regular font. Author name must be in 11 pt Regular font. Author affiliation must be in 10 pt Italic. Email address must be in 9 pt Courier Regular font.

TABLE I
Font Sizes for Papers

Font Size	Appearance (in Time New Roman or Times)		
	Regular	Bold	Italic
8	table caption (in Small Caps), figure caption, reference item		reference item (partial)
9	author email address (in Courier), cell in a table	abstract body	abstract heading (also in Bold)
10	level-1 heading (in Small Caps), paragraph		level-2 heading, level-3 heading, author affiliation
11	author name		
24	title		

- Follow the rules set by the format
  - Fonts, sizes, reference style etc.

Structure

- Title
  - Clearly explains what is done
- Abstract
  - Contains the main results and conclusions of the publication
- Introduction
  - What is the problem you are solving and why is it interesting?
- Methods
  - What are the methods, how do they work? Are there benchmark datasets? Can we create a taxonomy of the methods?
- Results
  - How do the methods compare in relation to your stated problem
  - Are there contradictions or gaps in the existing literature?
- Discussion / Conclusion
- References

How to write

- Review your literature research and notes
- Determine who your audience is
  - Your primary reader is the reviewer (me)
    - Think about what my concerns would be, address them early
- Create a first draft
  - Be creative, don't correct it until its done
    - It makes no sense to correct phrases you are deleting later

How to write

- Get started (the sooner the better)
- Create outline, high level structure of your paper
- Start with the easy parts
  - Introduction write it last
  - Start with methods, results, discussion, abstract

How to write

- Citations are essential in research papers
- Every method needs to be cited when it is first mentioned
  - *Method XYZ [1] allows to analyse….*
  - *[2] demonstrates that ….*
- If a figure is not made by you, the citation needs to be repated in the figure caption!

- All tables and figures present in the paper need to referenced in the text.
  - *As shown in figure 2…*
  - *Table 3 shows that…*

Research research

- How to do literature research and writing a review paper (or any paper) is an old problem
- If you are stuck, search the internet!
  - Blogs, videos etc.
  - There are even research papers on how to write them!

Next steps

How do we go from here

Brainstorm, group creation

- Write on a piece of paper keywords related to your research question
- If you already have a group, discuss research question and validate
- If you are alone, find matches with others using the keywords

Next step

- The next block concentrates on
  - Gathering sources
  - Grouping them
  - Reading and analyzing
- A small presentation will show your progress, your sources, tips and tricks in literature search, your problems etc.

Intermediate presentation

- Max 5 min presentation with 5 minutes discussion per group
  - Present your research question
    - Combination of ML technology and specific domain problem
  - First presentation of technology and current trends
    - Main sources, major methods, datasets?
  - Show your current open questions and issues
    - So we can discuss and help each other
  - Present first 30 sources you found
    - Quickly explain process, difficulties

